

Session Agent School - Product Spec

Separate module/product for creating, evaluating, improving and packaging musical Session Agents. Poppin Music is the studio; Session Agent School is the training room.

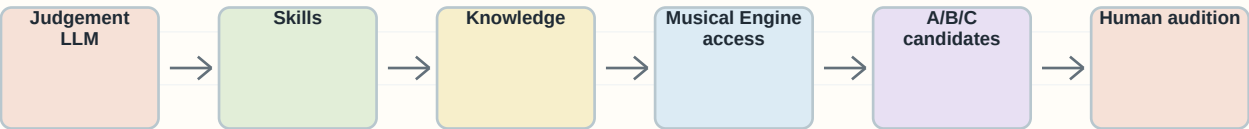
Product sentence	Session Agent School educates musical Session Agents through skills, knowledge, Musical Engine capabilities, automated validation, human audition selection, human reasons, versioning and promotion.
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1. Why School is separate

Poppin Music	Session Agent School
Creative studio: prompt, generate, audition, commit, edit in Reaper.	Training room: research, tests, evaluation records, Teacher LLM, versioning, promoted packages.
Should stay light and reliable.	Can run slower experiments without polluting live composition.
Consumes Session Agents.	Builds and improves Session Agents.

2. Agent anatomy School improves

Student agent anatomy



3. The School loop

Education loop



The Teacher LLM reasons. Deterministic systems verify. The human ear decides musical quality.

4. Teacher LLM responsibilities

#	Responsibility
1	Classify failure type.
2	Distinguish personal taste from general musicianship.
3	Inspect evaluation history.
4	Propose skill changes.
5	Propose new tests.
6	Identify missing deterministic tools.
7	Suggest when a music-native model might help.
8	Compare old vs new versions.
9	Prepare promotion notes.

5. Failure classes

Failure type	Meaning	Fix
Judgement	Technically valid but musically weak	Improve role instructions, examples, skills
Skill	Missing craft knowledge	Add/rewrite skill material
Knowledge	Weak reference/domain material	Add curated references or distilled notes
Engine	Correct idea rendered badly	Improve tools, MIDI rendering, voicing, validation
Representation	PMR cannot express the idea	Extend PMR schema
Model	Model unreliable for role	Test another model through gateway
Evaluation	Automated test rewards wrong thing	Fix evaluation design

6. Evaluation record

```
{
  "agent": "harmony",
  "agentVersion": "1.0.3",
  "scenarioId": "harmony_delayed_resolution_012",
  "prompt": "8 bars, peaceful but unresolved",
  "pmrContext": {},
  "candidates": ["A", "B", "C"],
  "selected": "C",
  "humanReason": "C keeps tension without sounding clever",
  "teacherDiagnosis": "needs stronger tension-with-simplicity skill",
  "hardValidation": "pass"
}
```

7. Evidence threshold and validation

Topic	Rule
One selection	Record only; do not rewrite the agent.
Repeated issue	Create targeted tests.
Repeated cross-context failure	Propose skill/engine change.
Old-vs-new wins repeatedly	Promote new version.
Hard validation	Schema, MIDI, range, timing, locks, bridge output.
Musical validation	Human listening: boring, clever, alive, restrained, responsive.

8. Agent package format

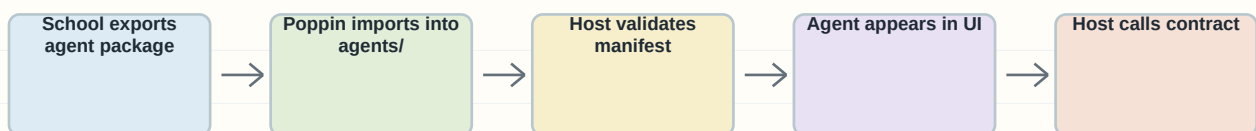
```
harmony-session-agent-v1.1/  
manifest.json  
README.md  
skills/  
prompts/  
evaluations/  
tools-required.json  
adapter.ts
```

Manifest shape

```
{  
  "id": "poppin.harmony",  
  "name": "Harmony Session Agent",  
  "version": "1.1.0",  
  "capabilities": ["harmony.generate", "harmony.reharmonize"],  
  "requiresPMR": ["tempo", "meter", "key", "sections", "harmony", "intent"],  
  "requiresTools": ["constructChord", "voiceLead", "renderHarmonyMidi"],  
  "outputSchema": "SessionAgentOutput.v1"  
}
```

9. How Poppin consumes School output

Import flow



10. Identity Memory boundary

Do not mix these

Session Agent Education asks what makes a competent expressive musician. Identity Memory observes what this particular user repeatedly chooses. Identity Memory must not silently steer new generation.

11. First School MVP

Step	MVP action
1	Manually build Harmony Session Agent v0.
2	Create 25 Harmony evaluation scenarios.
3	Generate A/B/C for each.
4	Human auditions and records selection + short reason.
5	Teacher LLM reviews records and proposes 3 skill improvements.
6	Apply one improvement manually.
7	Run old-vs-new blind comparison on 10 scenarios.
8	Promote only if clearly better.

12. North star

School succeeds when a Session Agent becomes less like a prompt and more like a musician: it listens, restrains itself, responds to context, generates alternatives with meaningful differences, explains honestly, and improves through actual audition rather than pretending that valid JSON is music.